

AI Health Navigator for Patients - System Overview

DESCRIPTION

The AI Health Navigator is an enterprise-scale AI platform offering real-time, patient-facing triage, guidance, and provider matching. Designed to integrate into healthcare portals or insurance apps, the platform supports secure, HIPAA-compliant symptom analysis and response delivery.

USE CASES

- Triage patients before ER visits
- Help users interpret insurance policies
- Recommend care level (urgent care vs. primary)
- Connect users to in-network providers

ARCHITECTURE

- Frontend: React + Flutter
- Backend: FastAPI microservices
- ML: LLM-based Q&A + classifier-based triage
- Database: PostgreSQL + Vector DB (FAISS)
- Infra: Docker + Kubernetes + Terraform
- CI/CD: GitHub Actions + Prometheus/Grafana

AI COMPONENTS

- ICD10/SNOMED-based symptom matcher

- LLM prompt chaining for nuanced Q&A
- Insurance policy retriever + summarizer
- Reinforcement learning for feedback-based triage tuning

SECURITY & COMPLIANCE

- Full PHI logging and encryption
- Role-based access control (RBAC)
- Audit trails for every API event
- HIPAA and GDPR alignment

SCALABILITY

- Auto-scaling Kubernetes clusters
- Millions of concurrent sessions
- Caching for common Q&A
- Streaming-friendly for mobile users